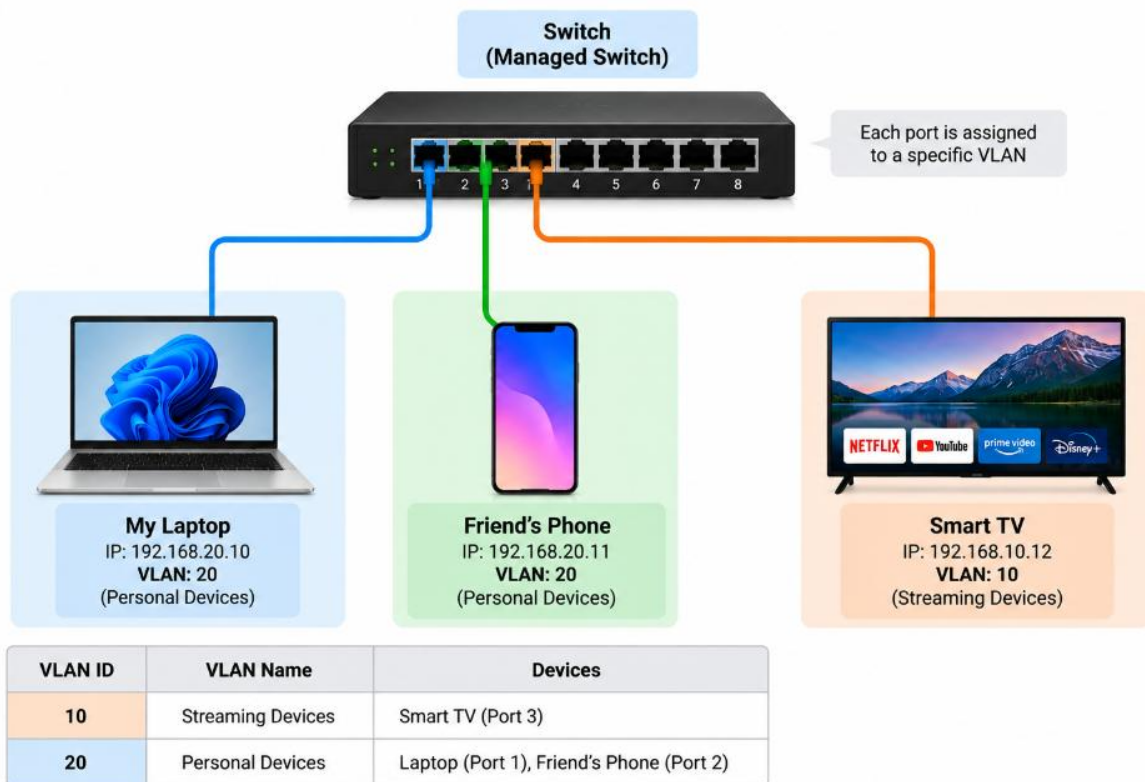


Switching & Vlan

Task – 1 : Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

Solution : **Network Diagram with VLANs**

A **VLAN (Virtual Local Area Network)** logically divides one physical switch into separate networks. Devices in different VLANs cannot communicate directly through a Layer 2 switch.



VLAN Assignment

Device	Switch Port	VLAN	Purpose
Smart TV	Fa0/1	VLAN 10	Streaming
Laptop	Fa0/2	VLAN 10	Streaming/Media
Another PC	Fa0/3	VLAN 10	Streaming
Phone	Fa0/4	VLAN 20	Personal

Device	Switch Port	VLAN	Purpose
Friend's Phone/PC	Fa0/5	VLAN 20	Personal

The switch keeps the two VLANs logically separated.

VLAN 10 devices can communicate with other VLAN 10 devices.

VLAN 20 devices can communicate with other VLAN 20 devices.

But:

VLAN 10 ➔ VLAN 20

unless a **router or Layer 3 switch** is configured for inter-VLAN routing.

Task – 2 : Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other. Hint : Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

Solution : **Create VLAN 10 and VLAN 20 in Packet Tracer**

Network Setup

PC1 — Fa0/1 ↙

PC2 — Fa0/2 |

PC3 — Fa0/3 |—— 2960 Switch

PC4 — Fa0/4 |

PC5 — Fa0/5 ↘

- **PC1–PC3 → VLAN 10 (STREAMING)**
- **PC4–PC5 → VLAN 20 (PERSONAL)**

Step 1: Create VLANs

enable

configure terminal

vlan 10

name STREAMING

exit

vlan 20

name PERSONAL

exit

Step 2: Assign Ports

VLAN 10:

```
interface range fa0/1 - 3
switchport mode access
switchport access vlan 10
exit
```

VLAN 20:

```
interface range fa0/4 - 5
switchport mode access
switchport access vlan 20
exit
```

Step 3: Save & Verify

```
end
copy running-config startup-config
show vlan brief
```

Expected:

VLAN 10 → Fa0/1, Fa0/2, Fa0/3

VLAN 20 → Fa0/4, Fa0/5

Step 4: Assign IPs

PC	VLAN	IP
PC1	10	192.168.10.10
PC2	10	192.168.10.11
PC3	10	192.168.10.12

PC	VLAN	IP
PC4	20	192.168.20.10
PC5	20	192.168.20.11

Subnet mask: 255.255.255.0

Step 5: Test

Same VLAN : PC1 → ping 192.168.10.11

Should work.

Different VLAN : PC1 → ping 192.168.20.10

Won't work because a Layer 2 switch **cannot route between VLANs**.

Task – 3 : In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

Solution : Move a PC from VLAN 10 to VLAN 20

Suppose **PC3** is initially connected to **Fa0/3**.

Before Configuration

PC3 is in VLAN 10.

```
interface fastEthernet 0/3
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
exit
```

Verify : show vlan brief

Test

PC3 : IP = 192.168.10.12

It should be able to ping another VLAN 10 PC : ping 192.168.10.10

Move PC3 to VLAN 20

Enter configuration mode:

```
enable
```

```
configure terminal
```

Then:

```
interface fastEthernet 0/3
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
exit
```

Save : end

copy running-config startup-config

After Configuration

Check : show vlan brief

Now it should show :

10 STREAMING active Fa0/1, Fa0/2

20 PERSONAL active Fa0/3, Fa0/4, Fa0/5

PC3 has now moved from VLAN 10 to VLAN 20.

Change Its IP Address

Because VLAN 20 is using the 192.168.20.0/24 subnet, PC3 should also use an address from that subnet.

For example:

PC3:

IP Address: 192.168.20.12

Subnet Mask: 255.255.255.0

Now test:

ping 192.168.20.10

PC3 should communicate with the VLAN 20 devices.

Task – 4 : Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3). Constraint : Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

Solution : **Assign 10 Ports to VLAN 30**

Instead of configuring each port individually, Cisco IOS supports the **interface range** command.

Suppose we want to assign **Fa0/1 through Fa0/10** to VLAN 30.

Configuration

enable

configure terminal

vlan 30

name GAMING

exit

interface range fastEthernet 0/1 - 10

switchport mode access

switchport access vlan 30

exit

end

copy running-config startup-config

Verify

show vlan brief

Explanation

Command	Purpose
vlan 30	Creates VLAN 30
name GAMING	Gives VLAN 30 a name
interface range fa0/1 - 10	Selects 10 ports at once
switchport mode access	Makes the ports access ports
switchport access vlan 30	Assigns the ports to VLAN 30
show vlan brief	Verifies VLAN and port assignments
copy running-config startup-config	Saves the configuration